

AI-Powered Image Editing System

Team 42 (Mid Prep 2-Adobe)

1 AI Relighting Engine: Condensed Market Analysis

Adobe's ecosystem—including Photoshop, Lightroom, Camera Raw, and Substance—offers a wide array of tools for exposure correction, tonal grading, masks, blend modes, and photo enhancement. Features such as Lighting Effects, Neural Filters, and Portrait Adjustments provide limited forms of lighting manipulation but rely strongly on preset lighting patterns or localized edits such as dodge/burn or highlight recovery.

While Adobe excels in color grading, selective adjustments, and tone mapping, traditional workflows treat a photograph as a static capture with limited flexibility to reposition light sources after the fact. Existing capabilities modify the final appearance rather than reconstruct illumination using scene geometry.

A gap persists between the precision lighting control available in Adobe's 3D tools and the ease-of-use expected in 2D photo editing.

Editing Workflow Comparison

Editing Operation	Traditional Adobe Workflow	AI-Enabled Workflow (Implemented)
Light direction adjustment	Requires layers, masks, feathering, and manual shadow shaping	Automatically adjusts shading based on predicted surface geometry
Shadow creation or modification	Achieved through brush work and blending modes	Dynamically generated as virtual light interacts with inferred depth
Highlight control	Requires detailed masking and tone curve adjustments	Emerges naturally from light manipulation
Realistic relighting	Time-intensive and artist-dependent	Achieved interactively in near real time

1.1 Rationale for an AI-Powered Relighting Workflow

Lighting is one of the most impactful creative decisions in photography, yet in traditional Adobe workflows it is mostly addressed during capture. The implemented system resolves this limitation by generating a pseudo-3D representation of the input image via depth and surface normal prediction.

The workflow enables users to:

- Place a virtual light source into the scene
- Move the light in X, Y, and Z space
- Adjust illumination direction
- Produce shading and shadow effects based on physical interaction

This bridges the gap between intuitive 2D editing interfaces and emerging geometry-aware AI tools, enabling lighting reconstruction rather than pixel-level correction.

1.2 Expected Creator Impact

Integrating spatial AI relighting into Adobe-style workflows has the potential to transform routine editing:

- **Creative flexibility:** Lighting becomes editable, enabling dramatic, soft, or directional looks without reshooting.
- **Efficiency:** Reduces dependence on brushes, masks, and layers.
- **Accessibility:** Enables advanced lighting effects without requiring users to learn 3D software.
- **Consistency:** Ideal for product photography, portraits, and advertising where consistent lighting is crucial.

2 AI Segmentation & Inpainting Engine: Condensed Market Analysis

2.1 Market Snapshot

The global AI visual editing ecosystem is expanding rapidly due to creator growth, mobile-first workflows, and rising demand for fast, accurate background removal. By 2025, AI photo editing represents a \$4.2B market, growing to \$14.5B by 2030 (28% CAGR). Mobile photo apps will expand from \$3.8B to \$10.2B (22% CAGR). Professional creative tools will grow from \$12.5B to \$18.4B (8% CAGR).

The total addressable market increases from \$20.5B in 2025 to \$43.1B in 2030 (19% CAGR).

2.2 Problem & User Gaps

Despite major investments in creative software, current tools remain inefficient, slow, and costly. Photoshop requires 5–15 minutes per object, includes a subscription cost, and has a steep learning curve. Remove.bg offers quick background removal but lacks object-level control and charges \$0.20–\$2.00 per image. Canva AI offers convenience, but limited accuracy.

User feedback shows:

- 68% report frustration with artifacts
- 52% struggle with selection tools
- 41% rely on multiple apps for a single task

The workflow is fragmented, time-consuming, and expensive.

Our engine addresses this through:

- 2–5 second segmentation
- 92–95% IoU accuracy
- Simple point-click UI
- API architecture for integration

2.3 Conclusion

A \$3–\$5B product gap exists between professional-grade software and low-accuracy, single-click apps. Our engine sits between these extremes:

- Prosumer AI editing
- Mobile-first design
- API-ready architecture

We enable segmentation and inpainting that is fast, accurate, affordable, and developer-friendly, making it well-positioned as market demand increases.